

A Symmetric Four-Leaf Tree Subcase of Basso's $\text{ae}(4)$ Conjecture

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Source Problem

The source paper is Giuliano Basso, *Absolute Lipschitz extendability and linear projection constants*, arXiv:2104.14238. Proposition 1.4 proves

$$\frac{5 + 4\sqrt{2}}{7} \leq \text{ae}(4) \leq \frac{3 + 6\sqrt{2}}{7},$$

and the introduction states that numerical simulations strongly suggest that the lower bound is sharp, i.e.

$$\text{ae}(4) = \frac{5 + 4\sqrt{2}}{7}.$$

The lower-bound example is a four-point rectangle in ℓ_∞^2 ; as a metric space it is the four leaves of a weighted tree with two internal vertices, four equal pendant edges, and a central edge of length $\sqrt{2} - 1$.

Result

For $r \geq 0$, let T_r be the weighted tree with vertices

$$p_1, p_2, q_1, q_2, u, v,$$

edges p_1u, p_2u, q_1v, q_2v of length 1, and central edge uv of length r . Let

$$X_r = \{p_1, p_2, q_1, q_2\}$$

with the metric induced from T_r .

Theorem 1.

$$\text{ae}(X_r) = \frac{(r+2)(r+3)}{r^2 + 3r + 4} \quad (r \geq 0).$$

Consequently

$$\sup_{r \geq 0} \text{ae}(X_r) = \frac{5 + 4\sqrt{2}}{7},$$

with equality at $r = \sqrt{2} - 1$.

Proof Intuition

Basso's theorem turns the absolute Lipschitz extension constant of a finite metric space into a relative projection constant of its Lipschitz-free space. For a tree this becomes especially concrete: the free space over the six vertices of T_r is ℓ_1^5 , one coordinate per edge. The four-leaf subspace is a three-dimensional “balanced flow” subspace. We compute its relative projection constant exactly by giving both sides of the projection constant duality: an explicit projection for the upper bound and an explicit trace witness for the lower bound.

Proof

For $r > 0$, the injective hull of X_r is the metric tree T_r . For $r = 0$, this is understood after identifying the two internal vertices; the formula below also follows by continuity, or directly from Basso's discrete n -pod computation. Let

$$Y_r = \{p_1, p_2, q_1, q_2, u, v\}.$$

Any Lipschitz map defined on Y_r extends to the whole tree by affine interpolation along the edges without increasing the Lipschitz constant. Hence Basso's reduction to the injective hull gives

$$\text{ae}(X_r) = \nu(X_r, Y_r) = \lambda(\mathcal{F}(X_r), \mathcal{F}(Y_r)).$$

Identify $\mathcal{F}(Y_r)$ with ℓ_1^5 by the edge basis, scaling the central edge coordinate by its length. We use coordinates (a, b, z, c, d) , where a, b are the two left pendant-edge coordinates, c, d the two right pendant-edge coordinates, and z the scaled central coordinate. The leaf subspace is

$$E_r = \mathcal{F}(X_r) = \left\{ (a, b, z, c, d) : a + b + c + d = 0, \quad z = \frac{r}{2}(a + b - c - d) \right\} \subset \ell_1^5.$$

Thus it remains to compute $\lambda(E_r, \ell_1^5)$.

Put

$$D = r^2 + 3r + 4, \quad t_r = \frac{(r+2)(r+3)}{D}, \quad q = \frac{r+3}{2D}.$$

For $x = (a, b, z, c, d) \in \ell_1^5$, define

$$s = a + b + c + d, \quad h = z - \frac{r}{2}(a + b - c - d),$$

and set $Q_r x = (a', b', z', c', d')$, where

$$\begin{aligned} a' &= a - \frac{s}{4} + qh, & b' &= b - \frac{s}{4} + qh, \\ c' &= c - \frac{s}{4} - qh, & d' &= d - \frac{s}{4} - qh, \\ z' &= z + (2rq - 1)h. \end{aligned}$$

A direct calculation gives $a' + b' + c' + d' = 0$ and

$$z' = \frac{r}{2}(a' + b' - c' - d').$$

Also $Q_r x = x$ whenever $x \in E_r$. Hence Q_r is a projection from ℓ_1^5 onto E_r .

With respect to the standard basis of ℓ_1^5 , the column ℓ_1 -norms of Q_r are all equal to t_r . Indeed the relevant signs are fixed for $r \geq 0$, since $qr < 1/2$, and the first column has norm

$$\left(\frac{3}{4} - \frac{qr}{2}\right) + \left(\frac{1}{4} + \frac{qr}{2}\right) + \frac{r}{2}(1 - 2qr) + 2\left(\frac{1}{4} - \frac{qr}{2}\right) = \frac{3}{2} + \frac{r}{2} - q(r^2 + r) = t_r,$$

while the central column has norm $2q(r + 2) = t_r$; the remaining pendant columns are symmetric. Since the operator norm on ℓ_1^5 is the maximum column ℓ_1 -norm, $\lambda(E_r, \ell_1^5) \leq t_r$.

For the reverse inequality we use the standard trace duality for relative projection constants in finite-dimensional polyhedral spaces: if $A(E_r) \subset E_r$ and $\nu_1(A) \leq 1$, then every projection P from ℓ_1^5 onto E_r satisfies

$$\|P\| \geq \text{tr}(A|_{E_r}).$$

Here ν_1 denotes the nuclear norm as an operator on ℓ_1^5 , so $\nu_1(A) = \sum_i \max_j |a_{ij}|$.

Let

$$S = \begin{pmatrix} 1 & -1 & 1 & -1 & -1 \\ -1 & 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & -1 & -1 \\ -1 & -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & -1 & 1 \end{pmatrix}$$

and $A = \text{diag}(\alpha, \alpha, \beta, \alpha, \alpha)S$, where

$$\alpha = \frac{r+2}{2D}, \quad \beta = \frac{r(r+1)}{D}.$$

The row maxima sum to $4\alpha + \beta = 1$, hence $\nu_1(A) = 1$. On the basis

$$e_1 = (1, -1, 0, 0, 0), \quad e_2 = (0, 0, 0, 1, -1), \quad e_3 = (1, 1, 2r, -1, -1)$$

of E_r , the matrix A acts diagonally:

$$Ae_1 = \frac{r+2}{D}e_1, \quad Ae_2 = \frac{r+2}{D}e_2, \quad Ae_3 = \frac{r^2+3r+2}{D}e_3.$$

In particular $A(E_r) \subset E_r$, and therefore

$$\lambda(E_r, \ell_1^5) \geq \text{tr}(A|_{E_r}) = \frac{r+2}{D} + \frac{r+2}{D} + \frac{r^2+3r+2}{D} = \frac{(r+2)(r+3)}{D}.$$

Together with the explicit projection this proves the formula for $\text{ae}(X_r)$.

Finally,

$$\frac{d}{dr} \frac{(r+2)(r+3)}{r^2+3r+4} = -\frac{2(r^2+2r-1)}{(r^2+3r+4)^2}.$$

The unique critical point on $[0, \infty)$ is $r = \sqrt{2} - 1$, where the maximum value is

$$\frac{5+4\sqrt{2}}{7}.$$

□

Verification Notes

The script `code/verify_symmetric_tree.py` evaluates the explicit projection and trace witness for representative values of r . It is only a sanity check; the proof above is symbolic.

The source paper itself contains the lower-bound example at Proposition 1.4 and proves the general reduction from absolute Lipschitz extendability to relative projection constants of Lipschitz-free spaces. Web and local-index searches on the phrases “four-leaf absolute Lipschitz extendability”, “two-cherry projection constant”, and the exact formula $(r+2)(r+3)/(r^2+3r+4)$ found no prior statement of this one-parameter subfamily formula.

Limitations

This does not prove Basso’s full conjecture for all four-point metric spaces. It proves the conjectured value for the symmetric tree-like family containing the proposed extremal example. The general four-point injective hull may have a genuine two-dimensional rectangular cell or unequal pendant lengths; those cases remain outside this packet.

References

- Giuliano Basso, *Absolute Lipschitz extendability and linear projection constants*, arXiv:2104.14238.
- The proof uses the reduction and trace-duality framework stated in the source paper, especially Theorem 1.2, Lemma 2.6, and Proposition 4.2 there.